SOFTWARE REQUIREMENTS SPECIFICATION

Sidekick PA

Your Personal, Proactive AI Companion

A private, offline-first desktop assistant for Windows that helps you manage reminders, routines and responsibilities, and speaks up when something needs your attention.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines what **Sidekick PA** shall do, how well it shall do it, and how a buyer can check that it does. It describes **Release 0.1, the Core Desktop Foundation**, in full detail and summarises the later releases so that the product direction is clear. It is written to the structure of IEEE/ISO/IEC 29148 so that a buyer, a developer and a tester can all work from the same document.

1.2 Product scope and value

Sidekick PA is a personal, proactive assistant for Windows. It lives on the user's computer as a small floating assistant and a full dashboard, keeps track of reminders, routines and responsibilities, and speaks up when something needs attention. Most productivity tools wait to be opened. Sidekick does not.

What a buyer gets.

- **Proactive help:** break prompts, reminders and renewal heads-ups arrive on their own, with the right priority and at a sensible moment.
- **Private and offline:** all data stays on the computer. Every Release 0.1 function works with no internet connection.
- **One assistant, not six apps:** the widget, chat and dashboard are three views of the same assistant, sharing one state.
- **Calm by design:** Focus mode and quiet hours keep the assistant quiet when the user needs it to be.
- **A clear growth path:** planner, finance, learning intelligence, voice, local AI and multi-device support build on the same foundation.

Out of scope for Release 0.1: voice input, finance, learning intelligence, machine learning, generative AI, cloud synchronization, Android and institutional (school) deployment. They are specified in section 5 so they can be planned, not built now.

1.3 Definitions and abbreviations

Terms used in this document are defined in Appendix A. Requirement priorities follow section 1.5.

1.4 References

- Sidekick PA Project Charter, Product Requirements and Technical Development Specification (the "Project Specification").
- Sidekick PA Product Vision and Visual Product Direction documents.
- Sidekick PA clickable prototype and visual design (identity, hybrid desktop presence and nine dashboard screens).
- ISO/IEC/IEEE 29148:2018, Systems and software engineering, Life cycle processes, Requirements engineering.
- The Data Protection Act, 2019 (Kenya).

1.5 Conventions

Each requirement has a unique ID and uses "shall" for obligation and "should" for a strong recommendation. Priorities are: **Must**, required for Release 0.1 to be accepted; **Should**, important and included unless risk or time forces a deferral that the buyer agrees to; **Could**, desirable. IDs are grouped by area (for example LC for lifecycle, RM for reminders, NT for notifications). Later-release requirements carry the prefix P followed by the phase number.

1.6 Document overview

Section 2 describes the product and its context. Section 3 gives five models: context, use cases, data flow, architecture and data. Section 4 lists the Release 0.1 requirements. Section 5 summarises later releases. Sections 6 to 8 cover interfaces, data and quality attributes. Sections 9 and 10 explain how the product is verified and how requirements trace back to the Project Specification. Sections 11 and 12 list risks and open issues.

2. Product Overview

2.1 Product perspective

Sidekick PA is a self-contained desktop product. It needs no server. It relies on Windows for notifications, the tray, start-up and an idle-time signal, and it stores everything in a local database. Figure 1 shows the system and its surroundings.

2.2 Product functions

Area	What it does	Release
Desktop presence	Floating assistant, tray, dashboard, light and dark themes	0.1
Reminders and scheduling	Create, complete and recur reminders; survive restarts and sleep	0.1
Notification engine	Priority, snooze, de-duplication, quiet hours, Focus mode, history	0.1
Productivity	Estimated active time, break prompts, focus sessions, summaries	0.1
Wellness reminders	Hydration, movement and eye-rest reminders	0.1
Conversational assistant	Offline text chat with confirmation before any change	0.1
Settings and data	Preferences, local storage, backup, integrity check	0.1
Planner and finance	Tasks, calendar, subscriptions, bills, savings goals (KES)	Phase 3
Learning intelligence	Subjects, quizzes, weak-topic detection, revision plans	Phase 4
Voice and local AI	Speech, natural language, local language model	Phase 5
Machine learning	Predictions from the user's own history	Phase 6
Multi-device and schools	Android, sync, profiles, institutional deployment	Phase 7

2.3 Users and stakeholders

Stakeholder	Interest
Primary user	One person using their own Windows laptop for study, development, personal projects and everyday routines. Release 0.1 is built for this user.
Buyer / product owner	Wants a dependable, private assistant with a clear roadmap and verifiable acceptance criteria.
Future users	University and secondary-school students, tutors, remote workers and freelancers.
Future institutions	Schools with unreliable internet. They need a separate institutional design (Phase 7) and are not served by Release 0.1.

2.4 Operating environment

Windows 11 (primary) and Windows 10 where supported by the chosen Electron version, on ordinary laptops including machines with 8 GB of RAM. The computer must be switched on for live notifications to be delivered. No internet connection is needed.

2.5 Design and implementation constraints

- **Technology:** Electron, React, TypeScript, Node.js and SQLite (through better-sqlite3), as set out in the Project Specification.
- **Offline-first:** essential functions shall not depend on an internet connection or an external AI service.
- **Local data:** one local user profile and local storage; no automatic cloud upload in Release 0.1.
- **Security:** context isolation on, Node integration off in the renderer, explicit and validated IPC.
- **Resource use:** suitable for computers with limited memory and processing power.
- **Scope discipline:** development proceeds in phases, and each module is stable and tested before the next begins.

2.6 Assumptions and dependencies

- The first deployment is one Windows computer with one local user.
- Windows start-up integration works only if the user turns it on.
- Idle detection estimates activity. It does not prove that the user is studying or working productively.
- Subscription and bill details are entered by the user (Phase 3).
- Machine-learning features need enough reliable user data (Phase 6).
- Voice recognition is not offline until a local speech engine is added (Phase 5).

2.7 Release plan

Release / phase	Focus	Detail in this SRS
0.1 Core Desktop Foundation	Background operation, widget, tray, database, reminders, notifications, activity monitoring, basic chat	Section 4 (full)
Phase 2	Productivity and wellness completion: habits, reports	Section 5.1
Phase 3	Planner and finance	Section 5.2
Phase 4	Learning intelligence	Section 5.3
Phase 5	Conversational AI and voice	Section 5.4
Phase 6	Machine learning	Section 5.5
Phase 7	Multi-device and institutional expansion	Section 5.6

3. System Models

3.1 System context

The user talks to Sidekick and Sidekick talks back. Windows provides notifications, the tray, start-up and idle time. Everything is stored in a local database. Dashed lines are later releases.

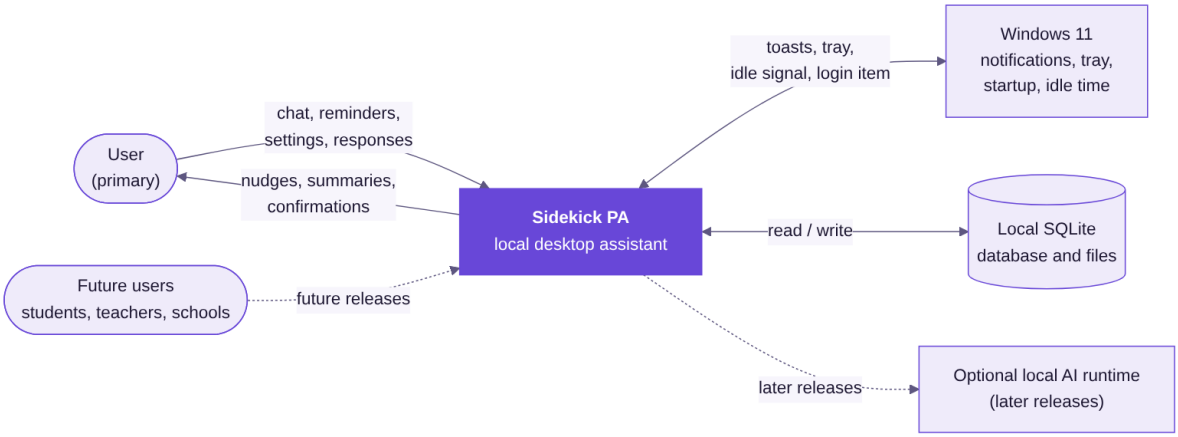


Figure 1. System context

3.2 Use cases

Release 0.1 has twelve use cases. The user starts nine of them. The scheduler and Windows take part in the other three, which Sidekick starts itself.

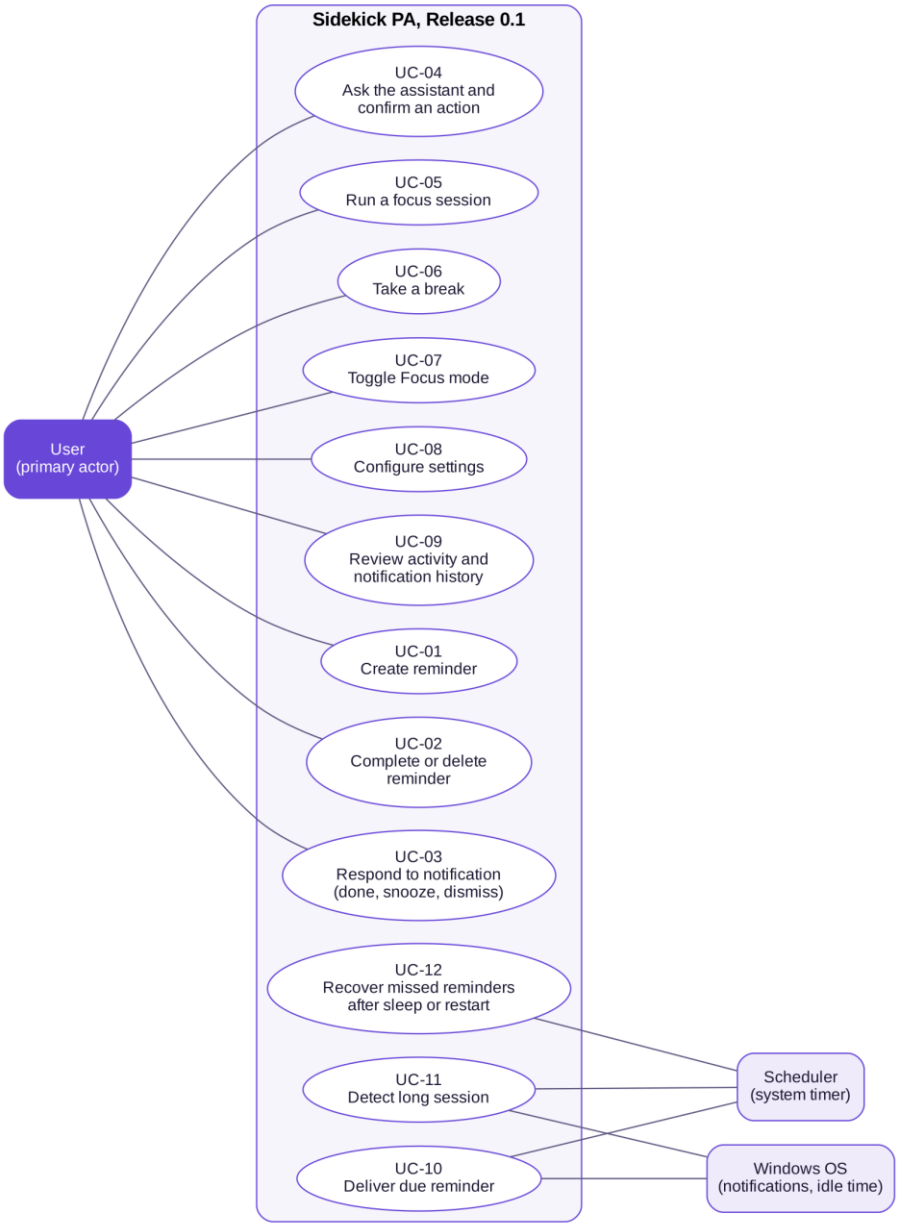


Figure 2. Use cases for Release 0.1

ID	Use case	Trigger and outcome	Requirements
UC-01	Create reminder	User adds a reminder in Planner, the widget or chat. It is saved and will fire at its due time.	RM-01, RM-06, RM-10
UC-02	Complete or delete reminder	User marks a reminder done or removes it. Recurring reminders roll forward.	RM-03, RM-04
UC-03	Respond to notification	User chooses Done, Snooze 10 min or Dismiss. The choice is recorded.	NT-03, NT-05, NT-06
UC-04	Ask the assistant	User types a request. Sidekick answers or asks for confirmation, then acts.	CH-01 to CH-05
UC-05	Run a focus session	User starts a 25, 50 or 90 minute session. Focus mode is on until it ends.	PR-07, NT-09
UC-06	Take a break	User answers a break prompt. The break is timed or the prompt is rescheduled.	PR-05, PR-06
UC-07	Toggle Focus mode	User switches Focus mode from the widget, tray, dashboard or chat.	NT-09, LC-11

ID	Use case	Trigger and outcome	Requirements
UC-08	Configure settings	User changes preferences. They apply at once.	ST-01, ST-02
UC-09	Review history	User views activity summaries and notification history.	PR-09, NT-11
UC-10	Deliver due reminder	A reminder reaches its due time. The notification engine applies policy and delivers it.	RM-08, NT-01 to NT-09
UC-11	Detect long session	Continuous activity reaches the threshold. A break prompt is raised.	PR-01, PR-04
UC-12	Recover missed reminders	After restart or sleep, overdue reminders are found and delivered once, as missed.	RM-09

3.3 Data flow

Figure 3 shows the main flows. All alerts pass through process 4.0, which reads the user's Focus mode and quiet-hours settings before anything is shown.

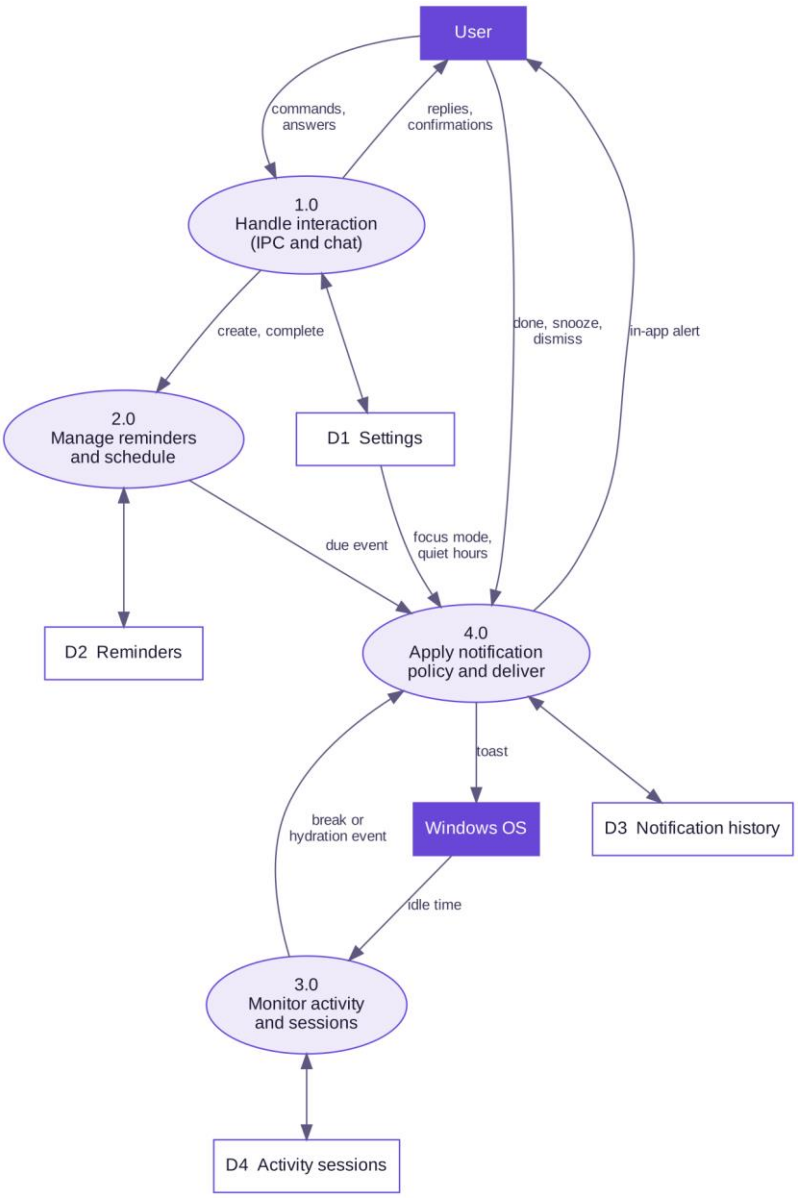


Figure 3. Level 1 data-flow diagram

3.4 Architecture layers

Sidekick uses a modular desktop architecture. The interface never touches the database or the operating system directly: it talks to the main process through a small, explicit and validated bridge. Modules do not call each other. They communicate through the assistant core, and every alert goes through the notification service.

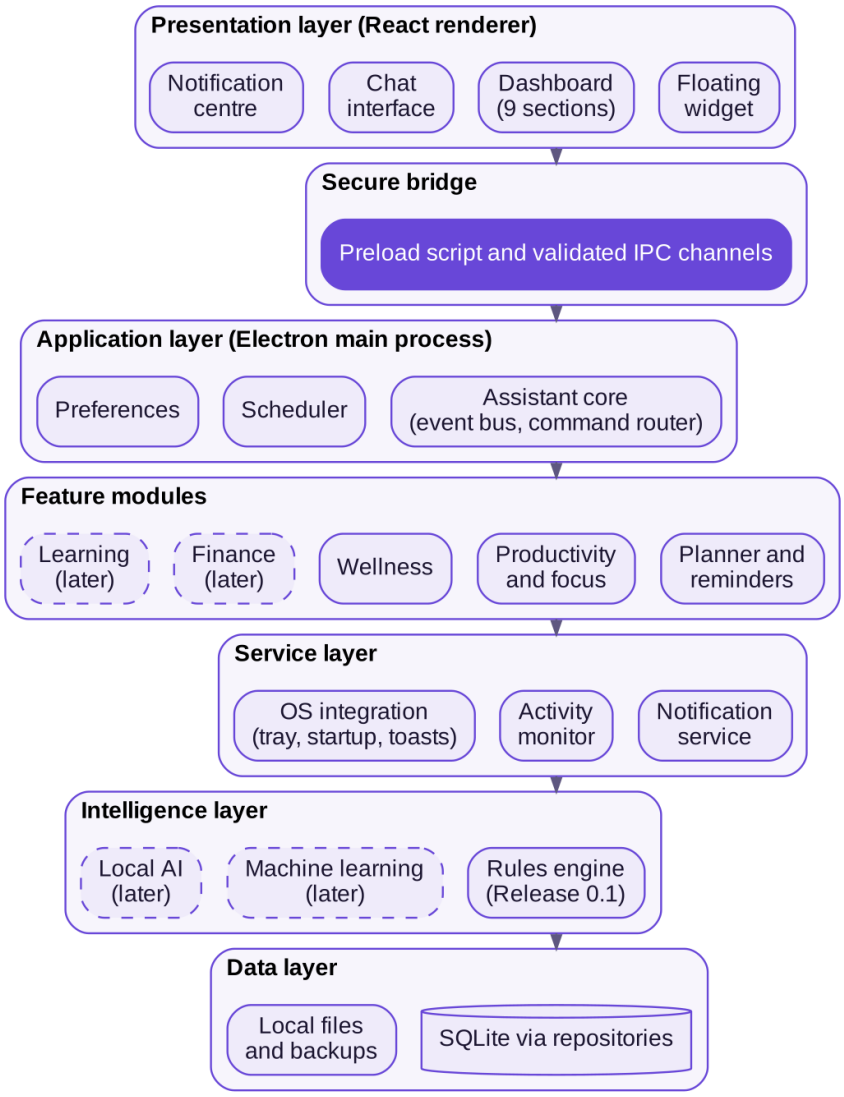


Figure 4. Architecture layers

3.5 Data model

Figure 5 shows the logical data model. Release 0.1 uses the five tables at the top. Later tables are shown so that early decisions leave room for them. Section 7 gives the data rules.

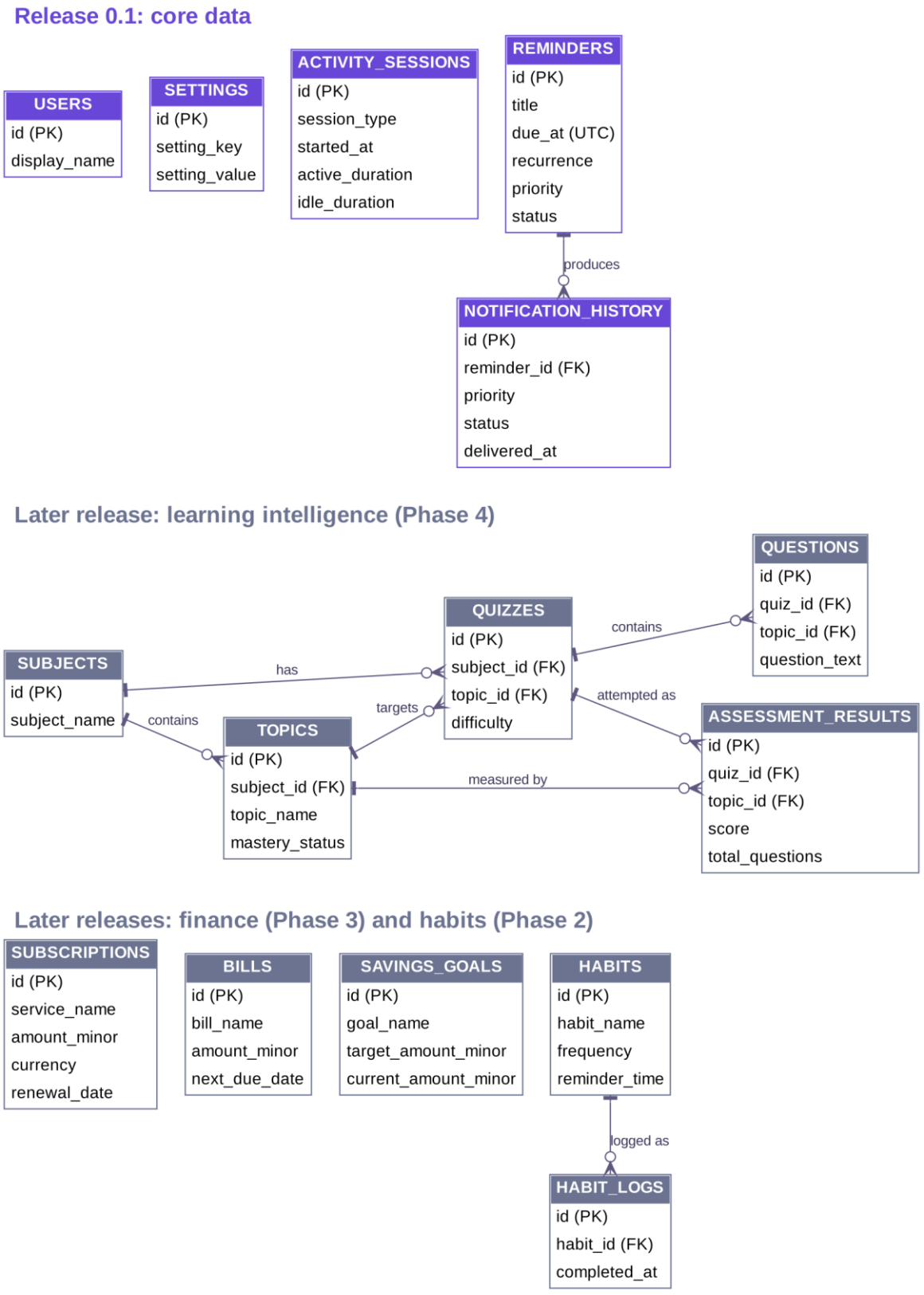


Figure 5. Logical data model (entity-relationship diagram)

4. Release 0.1 Requirements

These requirements define Release 0.1, the Core Desktop Foundation. All of them must be met for the release to be accepted, except where marked Should or Could.

4.1 Lifecycle and desktop presence

ID	Requirement	Priority
LC-01	Sidekick shall run as a desktop application on Windows 11 and be installed and launched like an ordinary Windows program.	Must
LC-02	Only one instance of Sidekick shall run at a time. Starting a second instance shall bring the existing one to the front.	Must
LC-03	Closing the dashboard window shall hide it and not exit the application. Background services (scheduler, activity monitor, notifications) shall continue to run. The application shall exit only when the user chooses Quit from the tray menu.	Must
LC-04	Sidekick shall show a system tray icon. Clicking it shall reopen the dashboard. The tray menu shall offer Open dashboard, Focus mode on/off, Pause reminders and Quit.	Must
LC-05	Sidekick shall provide a small floating widget that stays above ordinary windows and shows a time-of-day greeting, the next task, the number of items needing attention, a Focus mode toggle, an Ask Sidekick button and an Open dashboard link.	Must
LC-06	The floating assistant shall have three states the user can move between: a collapsed orb, the compact widget, and an expanded chat panel. The user shall be able to minimise, hide and restore it.	Must
LC-07	The widget shall not obstruct normal work: it shall be draggable, remain within the visible screen area, and remember its position between sessions.	Should
LC-08	The dashboard shall provide navigation to nine sections: Overview, Productivity, Learning, Finance, Wellness, Planner, Notifications, Chat and Settings. A section whose module is not part of the installed release shall show a clear explanatory state, and shall never present sample content as the user's own data.	Must
LC-09	Sidekick shall offer an opt-in setting to start automatically when the user signs in to Windows. The default shall be off.	Should
LC-10	Sidekick shall support a Dark theme and a Light theme, applied consistently to the widget, chat, dashboard and notifications. It should also offer Follow Windows appearance.	Must
LC-11	All windows shall present the same state. Focus mode, notification counts, reminders and settings shall have one source of truth in the main process, so a change in one window is reflected in the others without a restart.	Must

4.2 Reminders and scheduling

ID	Requirement	Priority
RM-01	The user shall be able to create a reminder with a title (required), description, category, due date and time, priority and recurrence.	Must
RM-02	The user shall be able to edit an existing reminder.	Should
RM-03	The user shall be able to mark a reminder complete. Completing a recurring reminder shall schedule its next occurrence.	Must
RM-04	The user shall be able to delete a reminder, after confirmation.	Must
RM-05	Sidekick shall list reminders by status and due date, including a Today view and an Overdue view.	Must

ID	Requirement	Priority
RM-06	Recurrence shall support none, daily, weekly and monthly. A monthly reminder set for a day that does not exist in a given month (for example the 31st) shall fall on the last day of that month.	Must
RM-07	Reminders shall be stored in the local database and shall survive application restarts and Windows restarts.	Must
RM-08	A due reminder shall fire while the application is running in the background, whether or not the dashboard is open.	Must
RM-09	After an application restart, Windows sleep and resume, or a system clock change, Sidekick shall detect reminders that came due in the meantime and deliver each at most once, marked as missed while the user was away. Several missed reminders shall be summarised and not delivered as a flood.	Must
RM-10	Each reminder shall carry one of four priorities: Critical, High, Normal or Low.	Must

4.3 Notification engine

ID	Requirement	Priority
NT-01	Every module shall raise alerts through one notification engine. No module shall show an alert on its own.	Must
NT-02	The engine shall assign each notification one of four priorities: Critical (important deadlines, user-defined urgent events), High (upcoming payments, important academic events), Normal (scheduled tasks, study reminders) and Low (hydration, movement and general wellness suggestions).	Must
NT-03	Notifications shall be delivered through the Windows notification system. Where Windows supports it, each shall offer Snooze 10 min, Done and Dismiss actions.	Must
NT-04	The dashboard shall include a notification centre that lists notifications, filters them by priority, and updates its counts live.	Must
NT-05	Snoozing a notification shall re-deliver it after 10 minutes.	Must
NT-06	Done, Snooze and Dismiss shall each be recorded with a timestamp.	Must
NT-07	The engine shall not deliver a duplicate of a notification (same source and subject) while an earlier one is still unresolved within a suppression window.	Must
NT-08	The user shall be able to define quiet hours (default 10:30 PM to 7:30 AM). During quiet hours, only Critical notifications shall be delivered, and only if the setting Allow critical alerts is on (default on).	Must
NT-09	While Focus mode is on, only Critical notifications shall be delivered, subject to the same Allow critical alerts setting. All others shall be held.	Must
NT-10	Held notifications shall stay visible in the notification centre and be summarised when quiet hours or Focus mode end. Held Low-priority notifications that are no longer relevant may be dropped and recorded as suppressed.	Should
NT-11	Notification history shall be kept and viewable, including counts of completed, postponed and dismissed items.	Must
NT-12	The engine shall limit overload by capping simultaneous toasts and batching Low-priority notifications. Exact limits are an open issue (OI-2).	Should
NT-13	A failed delivery shall be logged and the notification shall remain in the notification centre.	Should
NT-14	Notification wording shall be friendly and supportive, offer a choice of action, and never shame or pressure the user.	Should

Appendix B summarises how each priority behaves in normal use, quiet hours and Focus mode.

4.4 Productivity and activity monitoring

ID	Requirement	Priority
PR-01	Sidekick shall estimate active and idle computer time using the operating system's idle-time signal only. It shall not capture keystrokes, screenshots, window contents or application names.	Must
PR-02	The idle threshold, after which the user is considered idle, shall be configurable.	Must
PR-03	Sidekick shall record activity sessions and maintain estimated active time for the current day and week.	Must
PR-04	When continuous active time reaches the break threshold (default 90 minutes, configurable), Sidekick shall raise a break prompt such as "You've been at it for a while. Fancy a break?".	Must
PR-05	The break prompt shall offer Take a break, Remind me in 10 minutes and Dismiss. Each choice shall be recorded and shall take effect.	Must
PR-06	When the user takes a break, Sidekick shall let them know when the break length (default 10 minutes) is over.	Should
PR-07	The user shall be able to start and end a focus session of 25, 50 or 90 minutes, and be notified when it ends. Focus sessions shall switch Focus mode on for their duration.	Must
PR-08	The user shall be able to set working hours. Break prompts and activity estimates shall apply within them.	Should
PR-09	Sidekick shall show daily and weekly summaries of estimated active time, idle time, sessions and breaks.	Should
PR-10	Wherever activity is shown, Sidekick shall state that it is estimated from computer idle time and does not measure productivity.	Must
PR-11	The user shall be able to pause activity monitoring at any time.	Should

4.5 Wellness reminders

ID	Requirement	Priority
WL-01	The user shall be able to turn hydration reminders on or off and set their interval.	Must
WL-02	Sidekick shall offer movement reminders, triggered by a configured interval or by long continuous activity.	Should
WL-03	Sidekick shall offer eye-rest reminders at a configurable interval.	Should
WL-04	Wellness reminders shall have Low priority and shall be worded as suggestions. Sidekick shall never claim to detect hydration, fatigue or any medical condition.	Must
WL-05	Each wellness reminder type shall be individually adjustable and can be switched off.	Must

4.6 Conversational assistant

ID	Requirement	Priority
CH-01	Sidekick shall provide text chat in the expanded floating assistant and in the dashboard Chat section, sharing one conversation.	Must
CH-02	Chat shall work fully offline using rule-based understanding. No network connection shall be needed.	Must
CH-03	Chat shall support the command intents listed in Appendix C.	Must

ID	Requirement	Priority
CH-04	Before changing anything (creating, completing or deleting a reminder, changing a setting, starting a session), Sidekick shall show a confirmation with Confirm, Change and Cancel, and shall act only after Confirm.	Must
CH-05	When it does not understand a message, Sidekick shall say so kindly, suggest what it can do, and take no action.	Must
CH-06	The chat shall offer suggestion chips for common requests.	Should
CH-07	Sidekick shall be able to start a conversation, for example to offer a break, in the floating assistant. The user shall be able to answer by tapping a reply.	Must
CH-08	Conversations shall be stored locally and the user shall be able to view and delete them.	Should
CH-09	Voice input is not part of Release 0.1. The microphone control shall be absent, or shown disabled with the label "Voice is coming later". No microphone access shall be requested.	Should

4.7 Settings, data and privacy controls

ID	Requirement	Priority
ST-01	The Settings section shall let the user change: theme, start with Windows, quiet hours, Allow critical alerts, break and hydration intervals, idle threshold and working hours.	Must
ST-02	Settings shall be saved and applied immediately, without restarting the application.	Must
ST-03	All user data shall be stored locally. Release 0.1 shall make no automatic cloud upload and shall need no internet connection for any function.	Must
ST-04	The user shall be able to make a manual backup of their data to a location they choose.	Should
ST-05	The user shall be able to run a database integrity check. If a problem is found, Sidekick shall tell the user plainly that their data is safe and offer to restore from a backup.	Should
ST-06	The user shall be able to export their reminders, notification history and activity records in an open format (CSV or JSON).	Could
ST-07	Settings shall include a panel stating what Sidekick never does: capture keystrokes, take screenshots, read private messages, record the microphone continuously, collect passwords or read unrelated personal files.	Should
ST-08	On first run Sidekick shall ask for a display name and create the local user profile.	Should

5. Later-Release Requirements

These are planned capabilities, written at a lighter level than Release 0.1. They will be detailed before each phase begins. Priorities are provisional.

5.1 Productivity and wellness completion (Phase 2)

ID	Planned requirement	Priority	Phase
P2-01	Custom focus and study-session lengths, with sessions tagged as work or study.	Should	P2
P2-02	Custom recurring habits with completion tracking and a habit log.	Must	P2
P2-03	Sleep-routine reminders.	Should	P2

ID	Planned requirement	Priority	Phase
P2-04	Daily habit summaries.	Should	P2
P2-05	Downloadable daily and weekly activity reports.	Could	P2
P2-06	Reminder timing that adapts to the user's own history, using transparent rules.	Could	P2

5.2 Planner and finance (Phase 3)

ID	Planned requirement	Priority	Phase
P3-01	Tasks with deadlines, priorities and completion tracking; academic deadlines and project milestones.	Must	P3
P3-02	Appointments and a personal calendar with a month view.	Must	P3
P3-03	Scheduled backup reminders, for example "back up project files at the end of the month".	Should	P3
P3-04	Register subscriptions and recurring bills with name, amount, currency (KES by default), billing frequency, renewal date and days-before-renewal to be reminded.	Must	P3
P3-05	Money shall be stored as whole minor units (cents), never as floating-point numbers.	Must	P3
P3-06	High-priority advance renewal notifications.	Must	P3
P3-07	Upcoming-payments view and expected monthly expense totals.	Must	P3
P3-08	Savings goals with monthly reminders.	Should	P3
P3-09	Manual payment confirmation, and a history of payments and reminders.	Must	P3
P3-10	Sidekick shall not connect to bank accounts or hold payment credentials. Users enter billing details themselves.	Must	P3

5.3 Learning intelligence (Phase 4)

ID	Planned requirement	Priority	Phase
P4-01	Register subjects and academic units, with topics beneath each.	Must	P4
P4-02	Record completed topics.	Should	P4
P4-03	Create and administer quizzes and store their questions.	Must	P4
P4-04	Store every assessment attempt. Historical results shall never be overwritten.	Must	P4
P4-05	Classify each topic as Not studied, Struggling, Improving or Mastered, using documented, transparent rules.	Must	P4
P4-06	Identify repeatedly weak topics and explain the finding in plain language.	Must	P4
P4-07	Recommend revision sessions and schedule spaced revision.	Must	P4
P4-08	Adjust question difficulty to the user's performance.	Should	P4
P4-09	Produce learning progress reports.	Should	P4

5.4 Conversational AI and voice (Phase 5)

ID	Planned requirement	Priority	Phase
P5-01	Understand natural-language requests beyond the fixed command set.	Must	P5
P5-02	Voice input that the user switches on deliberately, with a clearly visible microphone indicator.	Must	P5
P5-03	A path to offline speech recognition and speech synthesis.	Should	P5
P5-04	Integration with a local language model for answers, study plans and summaries.	Should	P5
P5-05	Optional external AI services, off by default, never required, and never sent personal data without the user's permission.	Should	P5
P5-06	Every action proposed by the assistant shall need user approval.	Must	P5

5.5 Machine learning (Phase 6)

ID	Planned requirement	Priority	Phase
P6-01	Collect and validate historical data, and train only when there is enough of it.	Must	P6
P6-02	Predict topics likely to cause difficulty, helpful reminder timing and personal productivity patterns.	Should	P6
P6-03	Evaluate model accuracy, version models, and fall back to rule-based behaviour if a model is disabled or performs poorly.	Must	P6
P6-04	Explain each recommendation to the user.	Should	P6

5.6 Multi-device and institutional expansion (Phase 7)

ID	Planned requirement	Priority	Phase
P7-01	An Android companion application.	Could	P7
P7-02	Optional, secure synchronisation between a user's devices.	Could	P7
P7-03	Multiple user profiles on one computer.	Could	P7
P7-04	Institutional deployment as a separate architecture with teacher and administrator roles, student privacy protections, local-network synchronisation and periodic internet synchronisation.	Could	P7
P7-05	Offline educational content and timetables for schools with limited connectivity.	Could	P7

A school deployment shall not simply copy the personal version and add student accounts. It needs its own architecture, permissions and privacy design (P7-04).

6. External Interface Requirements

6.1 User interface

The interface follows the approved visual direction: friendly, premium and calm. It has a dark and a light theme designed separately (not inverted), a soft violet accent used sparingly for primary actions and active navigation, and a glowing orb as the assistant's identity. The orb is abstract, never a character.

- **Layout:** a left sidebar with the nine sections, a top bar with greeting, Focus toggle, notification bell and theme switcher, and a card-based content area with generous spacing.

- **Voice of the assistant:** friendly, observant and respectful. It speaks naturally ("You've been at it for a while. Fancy a break?"), offers a choice, and never shames. It does not act as though it controls the user's time.
- **Honesty:** wherever activity is shown, the interface says it is an estimate. Sample data is never shown as the user's own.
- **Accessibility:** keyboard operation, visible focus, 4.5:1 text contrast, text no smaller than 12 px, and meaning never carried by colour alone (USA-04 to USA-06).
- **Screens:** Appendix D lists each screen and its release.

6.2 Software interfaces

Interface	Use
Windows notifications	Deliver toasts with Snooze, Done and Dismiss actions (NT-03).
Windows system tray	Persistent icon and menu (LC-04).
Windows start-up	Optional start when the user signs in (LC-09).
Operating-system idle time	The only activity signal Release 0.1 uses (PR-01).
Local SQLite database	All persistent data, with transactional writes and enforced foreign keys.
File system	User-chosen locations for backups and exports (ST-04, ST-06).

6.3 Communications interfaces

None are required. Release 0.1 makes no network calls. Any future network feature (optional external AI, synchronisation) shall be off by default and opt-in.

6.4 Internal interface between interface and main process

The renderer reaches the main process only through a preload bridge that exposes a fixed list of operations: reminders (list, create, complete, remove), notifications (list, act), settings (get, update), activity summary, chat (send, confirm), window control, and a subscription to push events. Every request is validated in the main process (SEC-03).

7. Data Requirements

7.1 Release 0.1 data

Table	Holds	Notes
users	Local user profile (display name)	One row in Release 0.1
settings	Preferences as key and value	Applied immediately (ST-02)
reminders	Title, description, category, due time, recurrence, priority, status	Survive restarts (RM-07)
notification_history	Every notification and what happened to it	Linked to its source reminder where there is one
activity_sessions	Session type, start, end, active and idle duration	Idle-time estimates only (PR-01)

7.2 Data rules

ID	Requirement
DAT-01	Every important record shall have a unique identifier.

ID	Requirement
DAT-02	Dates and times shall be stored in UTC and displayed in the user's local time zone (Africa/Nairobi by default).
DAT-03	Monetary values shall be stored as integer minor units, never floating point.
DAT-04	Foreign keys shall be enforced.
DAT-05	Historical records, such as assessment results and notification history, shall not be overwritten by later events.
DAT-06	Schema changes shall be applied by version-controlled migrations, each applied once and atomically.
DAT-07	Sensitive information shall not be stored unless a feature needs it.

8. Non-Functional Requirements

Numeric targets for performance and resource use have not been set (open issue OI-1). Until they are, those requirements are verified against a baseline measured during acceptance.

8.1 Performance and resource use

ID	Requirement	Verification
PERF-01	The background process shall use little CPU while the user is idle.	Profiling against a baseline recorded during acceptance
PERF-02	Memory use shall suit ordinary laptops, including machines with 8 GB of RAM.	Profiling during acceptance
PERF-03	The dashboard and widget shall open without a perceptible delay, and database queries shall stay responsive as data grows.	Demonstration and profiling

8.2 Reliability

ID	Requirement	Verification
REL-01	Reminders, settings and history shall persist across restarts and Windows restarts.	Test
REL-02	Database writes shall be transactional so that a crash cannot leave partial records.	Test
REL-03	Critical operations (scheduling, notification) shall not depend on any window being open.	Test
REL-04	Scheduled work shall be recoverable after restart and after sleep or resume (see RM-09).	Test
REL-05	A failure in one module shall not stop the others, and failures shall be logged.	Test and inspection

8.3 Security

ID	Requirement	Verification
SEC-01	Electron context isolation shall be enabled for every window.	Inspection
SEC-02	Node integration shall be disabled in every renderer.	Inspection
SEC-03	The renderer may use only an explicit list of IPC channels, and every request shall be validated in the main process.	Inspection and test

ID	Requirement	Verification
SEC-04	The application shall load no remote code or content in its windows.	Inspection
SEC-05	Backups shall be written only to a location the user chooses. Optional encrypted backup is planned for a later release.	Test
SEC-06	No personal information shall be passed to any AI service without the user's explicit permission.	Inspection

8.4 Privacy

ID	Requirement	Verification
PRV-01	Personal data shall be stored locally by default.	Inspection
PRV-02	Sidekick shall not capture keystrokes or screenshots, monitor private messages, record the microphone continuously, collect passwords or read unrelated personal files.	Inspection and test
PRV-03	The design shall be consistent with the principles of the Data Protection Act, 2019 (Kenya): data minimisation, purpose limitation, security safeguards, and the user's rights of access, correction and erasure.	Inspection
PRV-04	The user shall be able to export their data and delete all of their local data.	Demonstration
PRV-05	Microphone access, when introduced, shall be user-activated and clearly indicated whenever active.	Inspection

8.5 Usability and accessibility

ID	Requirement	Verification
USA-01	Wording shall be friendly and supportive in the voice defined in section 6.1.	Inspection
USA-02	Frequent actions (answering a notification, toggling Focus mode, opening chat) shall be reachable in no more than two interactions.	Demonstration
USA-03	The assistant shall not overwhelm the user: no unprompted interruption in Focus mode or quiet hours beyond what NT-08 and NT-09 allow.	Test
USA-04	All controls shall be operable by keyboard and show a visible focus indicator.	Inspection
USA-05	Text shall meet at least a 4.5:1 contrast ratio in both themes, and no text shall be smaller than 12 px.	Inspection
USA-06	Priority and status shall never be conveyed by colour alone; an icon and a label shall accompany it.	Inspection

8.6 Compatibility

ID	Requirement	Verification
COM-01	Windows 11 shall be fully supported. Windows 10 shall be supported where the chosen Electron version supports it.	Test
COM-02	The interface shall work at common display scaling levels and screen sizes down to 1100 px wide for the dashboard.	Test

ID	Requirement	Verification
COM-03	Installing and running Sidekick should not need administrator rights.	Test

8.7 Maintainability

ID	Requirement	Verification
MNT-01	Each module shall be independently developed and testable, and modules shall communicate through defined interfaces or events, never through each other's internals.	Inspection
MNT-02	Business logic shall be separate from presentation. Database changes shall be made by version-controlled migrations.	Inspection
MNT-03	Reminder dates and recurrence, notification prioritisation, activity durations and settings validation shall have automated unit tests.	Test

8.8 Localisation

ID	Requirement	Verification
LOC-01	The interface shall be in English. Currency shall default to Kenya shillings (KES), dates shall be shown as day/month/year, and the time zone shall be Africa/Nairobi (UTC+3).	Inspection
LOC-02	Dates and times shall be stored in UTC and shown in the user's local time zone.	Test

9. Verification and Acceptance

9.1 Verification methods

Each requirement is verified by **inspection** (reading code, configuration or design), **demonstration** (operating the product), **test** (automated or scripted) or **analysis** (measurement, such as profiling). Every module is tested on its own before it is integrated.

9.2 Acceptance criteria for Release 0.1

AC-01 to AC-10 come from the Project Specification. AC-11 and AC-12 are added here because they protect the product's main promises.

ID	Release 0.1 is accepted when	Requirements
AC-01	The application launches successfully.	LC-01, LC-02
AC-02	The assistant keeps running after the dashboard is hidden.	LC-03
AC-03	The system tray can reopen the application.	LC-04
AC-04	User data persists after a restart.	RM-07, ST-03, REL-01
AC-05	Reminders can be created and completed.	RM-01, RM-03
AC-06	Due reminders produce desktop notifications.	RM-08, NT-03
AC-07	Break reminders operate at the configured threshold.	PR-04, PR-05
AC-08	Hydration reminders can be configured.	WL-01, WL-05
AC-09	The basic chat responds to the supported commands.	CH-02, CH-03, CH-04

ID	Release 0.1 is accepted when	Requirements
AC-10	No internet connection is needed for reminders and scheduling.	ST-03, RM-08
AC-11	Reminders that came due during sleep or while the app was closed are delivered once, as missed.	RM-09
AC-12	Quiet hours and Focus mode hold non-critical notifications and release them afterwards.	NT-08, NT-09, NT-10

9.3 User acceptance

The product owner uses Release 0.1 in ordinary daily work and judges whether reminders arrive at the right time, whether notifications are useful, whether the assistant gets in the way, whether activity estimates are reasonable, whether the product stays responsive, and whether data is still there after a restart.

10. Traceability to the Project Specification

Each functional requirement of the Project Specification (FR-01 to FR-20) maps to requirements in this SRS.

Spec ID	Specification requirement	SRS requirements	Release
FR-01	Launch as a desktop application	LC-01, LC-02	0.1
FR-02	Run in the background	LC-03, RM-08	0.1
FR-03	Display a floating assistant	LC-05, LC-06, LC-07	0.1
FR-04	Provide a system tray menu	LC-04	0.1
FR-05	Store data locally	RM-07, ST-03, DAT-01 to DAT-07	0.1
FR-06	Create and manage reminders	RM-01 to RM-10	0.1
FR-07	Send desktop notifications	NT-01 to NT-14	0.1
FR-08	Monitor idle and active time	PR-01 to PR-03	0.1
FR-09	Configure break reminders	PR-04 to PR-06	0.1
FR-10	Configure hydration reminders	WL-01 to WL-05	0.1
FR-11	Support conversational commands	CH-01 to CH-08	0.1
FR-12	Support voice interaction	CH-09 (excluded), P5-02, P5-03	Phase 5
FR-13	Manage subscriptions	P3-04 to P3-07, P3-10	Phase 3
FR-14	Manage recurring bills	P3-04 to P3-09	Phase 3
FR-15	Track learning progress	P4-01 to P4-04, P4-09	Phase 4
FR-16	Identify weak learning topics	P4-05 to P4-07	Phase 4
FR-17	Maintain habit history	P2-02, P2-04	Phase 2
FR-18	Predictive ML insights	P6-01 to P6-04	Phase 6
FR-19	Local generative AI	P5-04 to P5-06	Phase 5
FR-20	Synchronise across devices	P7-02	Phase 7

11. Risks

Risk	Mitigation in this SRS
Excessive RAM use	PERF-01, PERF-02: profile against a baseline; keep the background process lean.
Notification overload	NT-07 to NT-12, USA-03: priorities, de-duplication, quiet hours, Focus mode.
Missed reminders	RM-07 to RM-09, REL-04: persistent scheduling and recovery.
Inaccurate activity estimates	PR-01, PR-10: use idle time only and say so.
Internet-dependent voice	CH-09, P5-03: voice excluded from 0.1; local speech planned.
Poor ML predictions	P6-01, P6-03: train only on enough data, evaluate, fall back to rules.
Database corruption	REL-02, ST-04, ST-05: transactions, backup, integrity check.
Scope growth	Section 2.7 and section 5: strict phases; later work is specified but not built.
Privacy concerns	PRV-01 to PRV-05, SEC-06: local-first, explicit prohibitions.
Sleep or clock change breaks timers	RM-09: reconcile overdue events after resume.

12. Open Issues

These items are not yet decided. Nothing in this SRS assumes an answer to them.

ID	Issue	Proposed handling
OI-1	Numeric targets for CPU, memory, start-up time and reminder accuracy.	Measure a baseline during acceptance, then agree targets with the buyer.
OI-2	Default idle threshold, notification suppression window, and toast cap.	Propose defaults during Phase 2 design and confirm in user testing.
OI-3	Commercial model: licensing, price, support, maintenance and warranty.	Outside this SRS. Agree separately with the buyer.
OI-4	Installer, code signing and update mechanism.	Decide before the first external release.
OI-5	Minimum Windows 10 build supported by the chosen Electron version.	Confirm when the Electron version is fixed.
OI-6	Backup file format and encryption.	Plain backup in 0.1; encryption later.
OI-7	Default quiet hours (10:30 PM to 7:30 AM taken from the prototype).	Confirm with the primary user.
OI-8	How Learning and Finance sections appear in 0.1 (explanatory state or hidden).	Explanatory state per LC-08; confirm.
OI-9	How long to keep conversation history.	Keep until the user deletes it (CH-08); revisit.
OI-10	Interface languages beyond English, for example Kiswahili.	Not planned; revisit for school deployments.
OI-11	Whether selling or distributing Sidekick creates obligations under the Data Protection Act, 2019 (for example registration).	Take legal advice before commercial distribution. This SRS is not legal advice.

Appendix A. Glossary

Term	Meaning
Assistant core	The part of the main process that routes events and commands between modules.
Critical, High, Normal, Low	The four notification priorities (NT-02).
Electron	The framework that packages a web interface as a Windows desktop application.
Focus mode	A state in which only Critical notifications are delivered.
Idle time	Time since the user last used the keyboard or mouse, as reported by Windows. Sidekick never sees what was typed.
IPC	Inter-process communication between the interface and the main process.
KES	Kenya shilling.
Migration	A versioned change to the database structure.
Orb	The assistant's abstract visual identity.
Quiet hours	A daily period in which non-critical notifications are held.
Release 0.1	The Core Desktop Foundation milestone.
Snooze	Postpone a notification, by default 10 minutes.
SQLite	An embedded database stored in a single local file.
Toast	A Windows pop-up notification.
Widget	The compact floating assistant on the desktop.

Appendix B. Notification Behaviour

Priority	Typical sources	Normal use	Quiet hours	Focus mode
Critical	Important deadlines, user-defined urgent events	Toast and centre; stays until acted on	Delivered if Allow critical alerts is on	Delivered if Allow critical alerts is on
High	Upcoming payments, important academic events	Toast and centre	Held	Held
Normal	Scheduled tasks, study reminders	Toast and centre	Held	Held
Low	Hydration, movement, eye rest, general suggestions	Centre; toasts batched	Held or dropped if stale	Held or dropped if stale

Appendix C. Supported Chat Commands (Release 0.1)

Sidekick understands these intents by rules, offline. Anything else gets a friendly clarification (CH-05). Questions that need application-level data, such as "How much time did I spend coding?", are not supported, because Sidekick measures only computer idle time.

Intent	Example	Confirms first?	Result
Create reminder	Remind me to back up my project on Friday at 5 pm	Yes	Reminder saved

Intent	Example	Confirms first?	Result
Today's focus	What should I concentrate on today?	No	Today's and overdue reminders, by priority
Upcoming	What's coming up this week?	No	List of upcoming reminders
Complete reminder	Mark "Review notes" as done	Yes	Reminder completed
Focus mode	Turn on Focus mode	Yes	Focus mode switched
Start a break	Start a 10 minute break	Yes	Break timed
Start focus session	Start a 50 minute focus session	Yes	Session started
Activity today	How long have I been active today?	No	Estimated active time
Help	What can you do?	No	List of what Sidekick can do

Appendix D. Screen Inventory

Screen	Purpose	Release 0.1 content
Floating assistant	Quick access while working	Orb, widget, expanded chat (all)
Overview	The day at a glance	Schedule, tasks, activity, attention items
Productivity	Activity and focus	Active time, focus sessions, break prompt, summaries
Learning	Learning intelligence	Explanatory state (Phase 4)
Finance	Payments and savings	Explanatory state (Phase 3)
Wellness	Routine reminders	Hydration, movement, eye rest
Planner	Reminders and tasks	Reminders; tasks and calendar in Phase 3
Notifications	Everything Sidekick has told you	Centre, filters, actions, history
Chat	Talk to Sidekick	Text chat, confirmations
Settings	Preferences and privacy	All of ST-01 to ST-08